Thermodynamic description of the Mg–Si system: a synthetic benchmark document

ORCHID synthetic benchmark (generated; no human authors)

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Abstract

This synthetic document describes the Mg–Si binary system as calculated with pycalphad from the SGTE dataset MgSi-12Sch.tdb; every figure, table and stated value is generated, and the exact values are kept in a separate truth key. The highest invariant reaction, L → Mg2Si + (Si), lies at 939.3 °C.

1. Introduction

The Mg–Si system belongs to the 20 binaries of the ORCHID benchmark, chosen to cover diagram types such as congruent compound, simple eutectic. The thermodynamic description is the SGTE collection entry [1], which reproduces the assessment of M. Schick, B. Hallstedt, A. Glensk, B. Grabowski, T. Hickel, M. Hampl, J. Groebner, J. Neugebauer, R. Schmid-Fetzer, Calphad, 37, 77-86(2012). [2]. The calculations use pycalphad 0.11.2 at 1 atm.

The melting point of Mg reproduced by the dataset is 652.1 °C. Pure Si melts at 1413.8 °C.

2. Phase diagram and invariant reactions

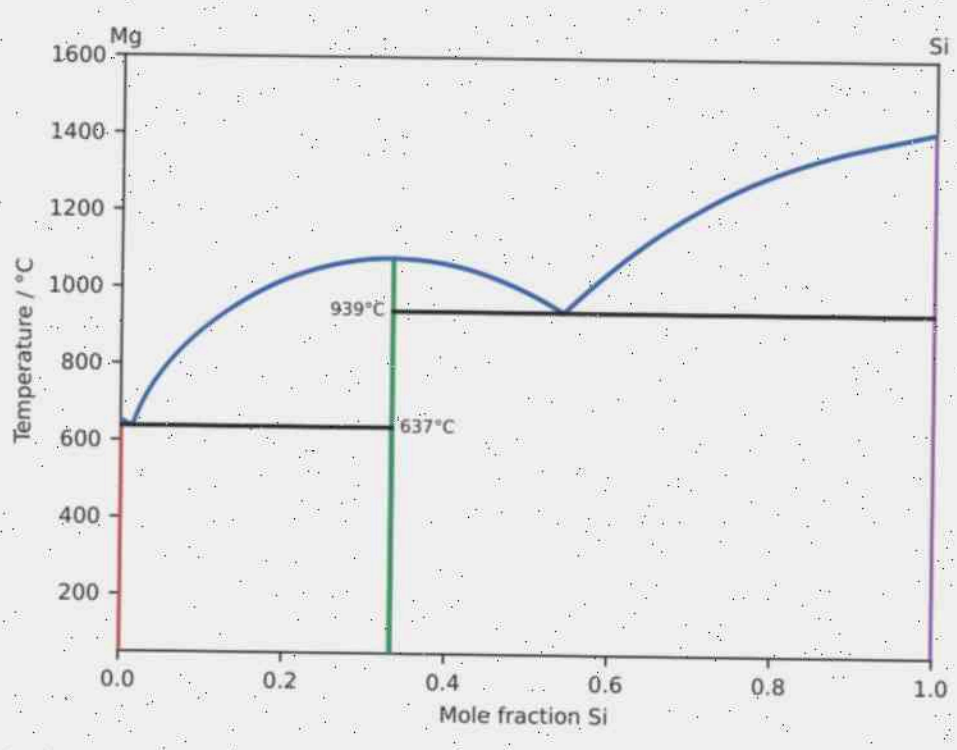


Fig. 1. Calculated Mg–Si phase diagram.

At 939.3 °C the eutectic L → Mg2Si + (Si) takes place; the L phase then contains 54.3 at.%. The eutectic reaction L → (Mg) + Mg2Si is calculated at 637.1 °C, with L at 1.5 at.% Si. Mg2Si melts congruently at 1077.4 °C and 33.3 at.% Si.

Table 1. Invariant reactions and congruent points, calculated (Mg–Si).

Reaction	Type	T (°C)	x(Si), 1	x(Si), 2	x(Si), 3
L → Mg2Si	congruent melting[a]	1077.4	0.333	0.333	–
L → Mg2Si + (Si)	eutectic	939.3	0.543	0.333	1.000
L → (Mg) + Mg2Si	eutectic	637.1	0.015	0.000	0.333

Phases 1–3 in the order written in the reaction. Calculated with pycalphad 0.11.2 from the SGTE dataset; synthetic document.
[a] Two-phase equilibrium at a temperature extremum; both phases have the same composition.

3. Phase stability

Table 2. Solid phases: structure, stability range and homogeneity range (Mg–Si).

Phase	Pearson	SB	Prototype	T min (K)	T max (K)	Range (x(Si))	T (K)
(Mg)	hP2	A3	Mg	< 300[a]	923.0	0.000	910
Mg2Si	cF12	C1	CaF2	< 300[a]	1350.5[b]	0.333	300
(Si)	cF8	A4	C (diamond)	< 300[a]	1687.0	1.000	300

Structure labels follow the TDB phase names (– where the name does not identify a structure).
Stability range: temperatures between which the phase is stable in some part of the diagram.
[a] Stable at the lowest calculated temperature.
[b] Congruent point (melting or transformation).

According to Table 2, Mg2Si remains stable up to 1077.4 °C (congruent melting).

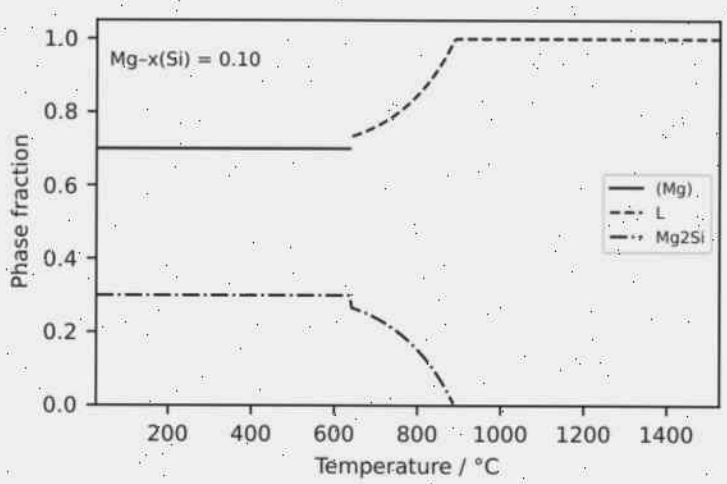


Fig. 2. Calculated equilibrium phase fractions versus temperature for Mg–x(Si) = 0.10.

4. Thermodynamic properties

At 1750 K the integral mixing enthalpy of the liquid has its minimum, −16345 J/mol, at 48.0 at.% Si. The calculated enthalpy of formation of Mg2Si at 298.15 K is −21.05 kJ/mol per mole of atoms. Its heat capacity at room temperature is 22.92 J/(mol K). For the equiatomic liquid at 1750 K, the Gibbs energy of mixing is −19.16 kJ/mol.

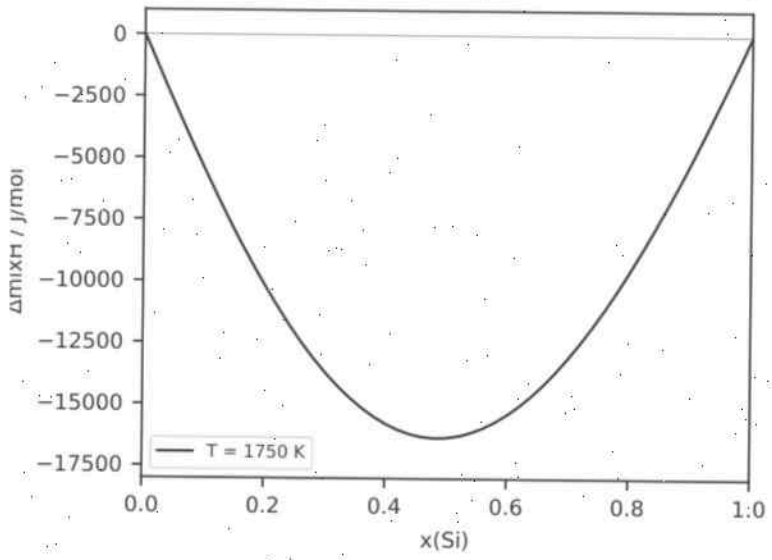


Fig. 3. Calculated mixing enthalpy of the liquid at 1750 K.

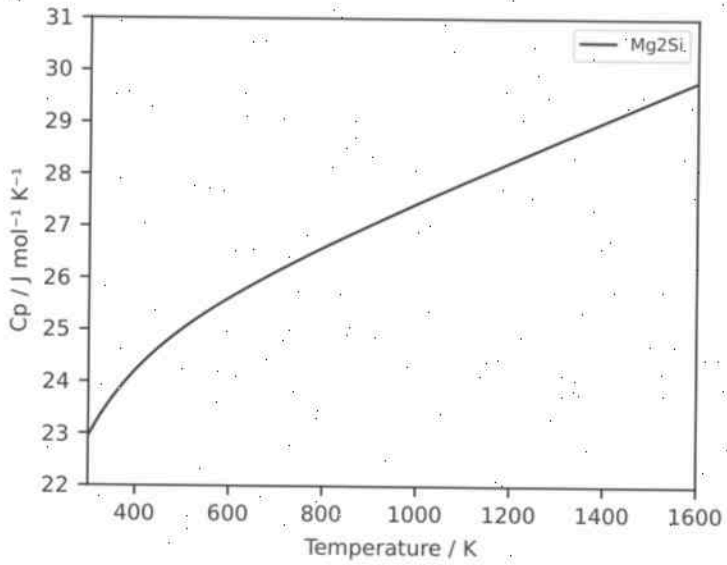


Fig. 4. Calculated heat capacity of Mg2Si.

Table 3. Calculated formation properties of the intermediate phases at 298.15 K (Mg–Si).

Phase	at.% Si	ΔfH(298.15 K) / kJ mol⁻¹ (f.u.)	ΔfS / J mol⁻¹ K⁻¹ (f.u.)	Cp(298.15 K) / J mol⁻¹ K⁻¹ (f.u.)
Mg2Si	33.3	−63.15 ± 2.0	−8.12	68.75

Reference: stable elements (SER) at 298.15 K; values per mole of formula units (formula unit as written in the Phase column). ± values are synthetic illustration uncertainties, not data.

Table 4. Integral and partial mixing properties of the liquid at 1750 K (Mg–Si).

at.% Si	ΔmixH / J mol⁻¹	ΔmixG / J mol⁻¹	a(Mg)	a(Si)
10	−5470	−8120	0.8764	0.0124
20	−10175	−13291	0.7196	0.0387
30	−13693	−16738	0.5493	0.0874
40	−15777	−18673	0.3858	0.1687
50	−16334	−19157	0.2476	0.2902
60	−15394	−18218	0.1453	0.4489
70	−13082	−15889	0.0789	0.6242
80	−9590	−12215	0.0399	0.7834
90	−5149	−7204	0.0175	0.9043

Reference states: pure liquid Mg and Si at 1750 K; per mole of atoms.

5. Conclusions

This document is synthetic and exists only to test extraction tools; its truth key lists every boundary, field, invariant, plotted point, table cell and stated value.

References

[1] B. Hallstedt, SGTE binary dataset collection, Calphad 89 (2025) 102833 (CC BY 4.0).

[2] M. Schick, B. Hallstedt, A. Glensk, B. Grabowski, T. Hickel, M. Hampl, J. Groebner, J. Neugebauer, R. Schmid-Fetzer, Calphad, 37, 77-86(2012).

[3] K. Synthetica, L. Exemplar, J. Fict. Phase Equilib. 12 (1987) 345–352 — fictitious reference, invented for this synthetic benchmark.

[4] M. Placeholder, Z. Notreal, Fict. Thermochim. Acta 41 (1994) 77–89 — fictitious reference, invented for this synthetic benchmark.

[5] R. Imaginary, Proc. Synth. Calorim. Soc. 8 (2003) 1102–1110 — fictitious reference, invented for this synthetic benchmark.